

- For the reactions that we study, there are some relatively predictable patterns that we see that we can then use to predict the products of a reaction
- We're going to go over 4 types of reactions (we'll talk about the 5th one that you did observe but weren't directed to make observations of tomorrow), what they're called, and which reactant combinations will most likely yield that type of reaction

- Reaction Type: Synthesis
 - Also sometimes called “combination” (term the book uses)
 - 2 or more substances react to form one new product
 - $A + B \rightarrow AB$
 - Ex: aluminum + oxygen $\rightarrow$ aluminum oxide
 - Any time you have 2 elements reacting, it will be synthesis; there are synthesis reactions between molecular compounds (we won't focus on those as much)

- Reaction Type: Decomposition
 - A single compound reacts, breaking down to form two or more products (elements or compounds)
 - Frequently, compound must be heated in order for reaction to occur
 - $XY \rightarrow X + Y$
 - Ex: potassium chlorate $\rightarrow$ potassium chloride + oxygen
 - There are further “rules”/patterns about decomposition that we will talk about tomorrow

- Reaction Type: Single Replacement
 - Also sometimes called “single displacement” (book uses replacement)
 - An element reacts with a compound to replace one of the elements in the compound, creating a new compound and leaving a new element
 - Frequently the compound is ionic, so the element replaces the element in the compound that forms the same type of charge
 - $AB + X \rightarrow XB + A$
 - Ex: aluminum nitrate + calcium $\rightarrow$ calcium nitrate + aluminum

- Reaction Type: Double Replacement
 - Also sometimes called “double displacement” (book uses replacement)
 - Two ionic compounds react, swapping cations to form two new ionic compounds
 - $AB + XY \rightarrow XB + AY$
 - Ex: barium chloride + sodium phosphate $\rightarrow$ barium phosphate + sodium chloride

- Determine Reaction Type Based on Reactants:
 - Two elements = synthesis
 - One compound = decomposition
 - One element + one compound = single replacement
 - Two compounds = double replacement
- To get balanced reaction equation:
 - Step 1: Use reactants to determine reaction type
 - Step 2: Predict products based on reactant type, write word equation
 - Step 3: Write skeleton equation based on word equation
 - Step 4: Balance skeleton equation with coefficients as needed